

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Technical Presentation

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO1 – Imitation (L1)	Assessment:	Assessment Tool 2 – Technical Presentation
Weightage:	30% of CIA		
CO1 Statement:	Analyse NLP models, regular expression patterns, finite state automata, morphological processing techniques and to interpret language processing. (BL-3)		
Student Name:		Reg. No.:	
Section / Batch:		Date:	

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- Suggested Team Member Roles (3 Members)
- Member 1: Theory, Concepts, and Literature Review
- Member 2: Algorithm Design, Models, and Technical Analysis
- Member 3: Demonstration, Case Study, Applications, and Conclusion
- GPS photo must submit.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
Evolution of Natural Language Processing: From Rule-Based Systems to Modern NLP	NLP Models, Language Processing Evolution	TP-01
Comparative Study of NLP Models and Algorithms	Model Analysis, Algorithm Comparison	TP-02
Regular Expressions for Real-World Text Processing	Pattern Matching, Information Extraction	TP-03
Designing Efficient Regular Expression Patterns for Information Extraction	Text Mining, Data Extraction Techniques	TP-04
Finite State Automata in Natural Language Processing	State Transitions, Language Recognition	TP-05
Building a Finite State Automaton for Token Recognition	Lexical Analysis, Token Validation	TP-06
Morphology and Word Formation in Natural Languages	Morpheme Analysis, Word Structure Processing	TP-07
Inflectional Morphology: Understanding Grammatical Variations	Grammatical Analysis, Word Normalization	TP-08
Derivational Morphology and Vocabulary Expansion	Word Generation, Semantic Transformation	TP-09
Comparative Analysis of Inflectional and Derivational Morphology	Morphological Classification, Language Analysis	TP-10

Finite-State Morphological Parsing Techniques	Morphological Parsing, Finite-State Processing	TP-11
Morphological Analysis for Indian Languages	Language-Specific Morphology, NLP Challenges	TP-12
Porter Stemmer: Design, Working Mechanism, and Applications	Stemming Algorithms, Text Preprocessing	TP-13
Stemming vs Morphological Parsing: A Comparative Study	Word Reduction, Morphological Processing	TP-14
Integrated NLP Pipeline: From Regular Expressions to Morphological Processing	End-to-End Language Processing Pipeline	TP-15

Annexure A – Technical Presentation

Team No.	Technical Presentation Title	Description
1	Evolution of Natural Language Processing: From Rule-Based Systems to Modern NLP	Analyze the historical development of NLP models, key milestones, and how language processing has evolved from handcrafted rules to intelligent systems.
2	Comparative Study of NLP Models and Algorithms	Explore different NLP models and algorithms used for language processing, comparing their working principles, strengths, and limitations.
3	Regular Expressions for Real-World Text Processing	Demonstrate how regular expressions are used for data validation, text extraction, log analysis, and information retrieval in practical applications.
4	Designing Efficient Regular Expression Patterns for Information Extraction	Analyze complex regular expression patterns used to extract emails, phone numbers, URLs, dates, and structured information from unstructured text.
5	Finite State Automata in Natural Language Processing	Explain the concepts of deterministic and non-deterministic finite automata and their role in lexical analysis and language recognition.
6	Building a Finite State Automaton for Token Recognition	Design and demonstrate a finite state automaton that recognizes tokens such as identifiers, keywords, and numbers in textual data.
7	Morphology and Word Formation in Natural Languages	Investigate how words are structured and formed using morphemes, and analyze the importance of morphology in NLP systems.
8	Inflectional Morphology: Understanding Grammatical Variations	Examine how inflectional changes affect word forms based on tense, number, gender, and case across different languages.
9	Derivational Morphology and Vocabulary Expansion	Analyze derivational processes that create new words and study their impact on semantic interpretation and language understanding.
10	Comparative Analysis of Inflectional and Derivational Morphology	Compare the characteristics, applications, and challenges of inflectional and derivational morphology in NLP tasks.
11	Finite-State Morphological Parsing Techniques	Explore finite-state transducers and parsing techniques used to analyze and generate morphological structures in natural languages.
12	Morphological Analysis for Indian Languages	Study the challenges and solutions involved in performing morphological processing for morphologically rich Indian languages such as Tamil, Hindi, and Telugu.
13	Porter Stemmer: Design, Working Mechanism, and Applications	Analyze the Porter Stemming algorithm, its rule-based approach, implementation steps, and role in information retrieval systems.
14	Stemming vs Morphological Parsing: A Comparative Study	Compare stemming and morphological parsing techniques based on accuracy, computational complexity, and application scenarios.
15	Integrated NLP Pipeline: From Regular Expressions to Morphological Processing	Demonstrate a complete NLP preprocessing pipeline incorporating regular expressions, finite state automata, stemming, and morphological analysis for text processing.

Rubrics for Calculation

Technical Presentation					
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Team No. / Criteria Level	Technical Content Accuracy	Problem Identification	Use of Tools	Communication	Professionalism	Total %

Faculty In-charge	Course Coordinator	Program Director
Dr.T.Kumaragurubaran		
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Knowledge Sources in Natural Language Processing (NLP)

Presented by: Y.Pooja Sri

Reg No : 192324143

What is NLP?

-
- Natural Language Processing (NLP) is a field of Artificial Intelligence (AI) that focuses on enabling computers to understand, interpret, and generate human language in both text and speech forms.
 - **NLP combines concepts from:**
 - Artificial Intelligence
 - Machine Learning
 - Computer Science
 - **Why are Knowledge Sources Needed?**
 - Human language is complex because it contains:
 - Different pronunciations
 - Word formations
 - Grammar rules
 - Multiple meanings
 - Context Conversations

Phonetics & Phonology

Definition

- **Phonetics** is the study of speech sounds and how they are produced, transmitted, and perceived.
- **Phonology** is the study of how speech sounds are organized and function within a particular language.
- Example: 'Cat' → /kæt/

Explanation:

The first knowledge source is Phonetics and Phonology. Phonetics studies how speech sounds are produced and pronounced. Phonology studies how these sounds are organized in a language. This knowledge helps systems like Siri and Google Assistant recognize spoken words.

Morphology

Definition: Morphology is the study of how words are formed and how their parts (such as prefixes, suffixes, and root words) affect their meaning.

Examples:

Play → Played

Happy → Happiness

Explanation:

Morphology studies how words are formed. It identifies the root word and any prefixes or suffixes attached to it. For example, 'Played' comes from the root word 'Play', and 'Happiness' comes from 'Happy'. This is useful in spell checking and machine translation.

Syntax

Definition : Syntax is the study of how words are arranged to form grammatically correct sentences.

Example:

Correct: The boy is playing.

Incorrect: Boy the playing is.

Explanation:

Syntax focuses on grammar and sentence structure. It checks whether words are arranged correctly to form a meaningful sentence. This knowledge is used in grammar-checking software and chatbots.

Semantics



Definition : Semantics is the study of the meaning of words, phrases, and sentences.

Example:

The cat sat on the mat.

Explanation:

Semantics helps the computer understand the meaning of words and sentences. For example, in the sentence 'The cat sat on the mat,' the system understands that the cat is performing the action of sitting.

Pragmatics

Definition : Pragmatics is the study of how the meaning of a sentence changes based on the context in which it is used.

Example:

Person A: It's cold.

Person B: I'll close the window.

Explanation:

Pragmatics goes beyond the literal meaning. It understands what the speaker actually intends based on the context. Here, 'It's cold' is understood as an indirect request to close the window.

Discourse

Definition : Discourse is the study of how sentences are connected to make a meaningful conversation or text.

Example:

- Rahul bought a laptop.
- He uses it every day.

Explanation:

Discourse studies how multiple sentences are connected. In this example, the system understands that 'He' refers to Rahul and 'it' refers to the laptop. This helps in text summarization and conversation analysis.

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Applications

- ChatGPT
- Google Translate
- Siri
- Alexa
- Speech Recognition
- Grammar Checker

Explanation:

All these knowledge sources are used in real-world NLP applications such as ChatGPT, Google Translate, Siri, Alexa, and grammar-checking tools.

Conclusion

- To conclude, Knowledge Sources in Natural Language Processing (NLP) help computers understand and process human language effectively. Each knowledge source has a specific role: Phonetics & Phonology deals with speech sounds, Morphology studies word formation, Syntax focuses on sentence structure, Semantics understands meaning, Pragmatics interprets meaning based on context, and Discourse connects sentences to understand the complete text or conversation. Together, these knowledge sources enable NLP applications such as chatbots, virtual assistants, machine translation, speech recognition, and text summarization. They form the foundation of NLP, allowing computers to communicate with humans in a more accurate and natural way.

THANK YOU

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 a &\rightarrow \hat{P}(a|b) = \frac{4}{8} - \frac{1}{8} = \frac{3}{8} \\
 b &\rightarrow \hat{P}(b|b) = \frac{3}{8} - \frac{1}{8} = \frac{2}{8} \\
 c &\rightarrow \lambda_b \hat{P}(c) \Rightarrow \lambda_b \left[\frac{9}{32} - \frac{1}{8} \right] = \lambda_b \frac{5}{32} \\
 d &\rightarrow \lambda_b \hat{P}(d) \Rightarrow \lambda_b \left[\frac{7}{32} - \frac{1}{8} \right] = \lambda_b \frac{3}{32}
 \end{aligned}$$

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